

AI POWERED THREAT DETECTION SYSTEM

College Viva & Professional Presentation Guide with Speaker Script

1. Presentation Strategy & Viva Overview

This guide equips you to present the **AI Powered Threat Detection System** during a final-year viva voce, academic evaluation, or professional technical presentation. It includes a slide-by-slide verbal presentation script, live demonstration steps, and defensible answers for anticipated examiner questions.

2. Slide-by-Slide Presentation Script

Slide # & Topic	Key Slide Visuals / Points	Presenter Script (What to Say)
Slide 1: Title & Overview	Project Title, Student Name, Guide Name, KSOU / University Logo	'Respected Examiners, good morning/afternoon. Today I present my final year project titled AI POWERED THREAT DETECTION SYSTEM. This system combines Deep Learning and Machine Learning to detect cyber-attacks in real time.'
Slide 2: Problem Statement	Traditional IDS weaknesses, high False Positive rate, lack of zero-day coverage	'Traditional signature-based IDS fail against evolving cyber attacks like DoS, DDoS, Port Scans, and ARP Spoofing. They generate excessive false alarms and lack mathematical explainability.'
Slide 3: Proposed Architecture	Dual AI Classifier Diagram: Deep MLP Flow Engine + Random Forest Protocol Engine	'Our proposed system introduces a dual AI architecture: A 3-layer Deep MLP Neural Network processes 12 flow features for volumetric attacks, while a Random Forest processes 5 ARP features for protocol tampering.'
Slide 4: Real-Time Results & Metrics	Flow Model: 99.75% Acc ARP Model: 100% Acc 14,000 Samples	'Our model was evaluated on 14,000 benchmark samples modeled after CICIDS2017 and UNSW-NB15 datasets, achieving 99.75% accuracy on flow traffic and 100% on protocol anomalies.'
Slide 5: MITRE & Active Mitigation	MITRE ATT&CK; Matrix (T1499, T1498, T1046, T1110, T1557) & Firewall Rule Generator	'Every attack is mapped to standard MITRE ATT&CK; techniques, and the system automatically generates actionable Linux IPTables and Fail2Ban firewall rules for instant threat blocking.'

3. Recommended Live Demonstration Flow

- Open the live application at <https://ai-powered-threat-detection-system.vercel.app> in Chrome.
- Point out the **Fixed Left Navigation Bar** and explain how it locks in place when scrolling.
- Highlight the **Real-Time Threat Score Timeline** chart showing a steady flat line at 5.0 during normal stream.
- Click the **DoS Attack (T1499)** simulator button at the top. Show the graph spike to 83.7 / High Risk immediately below the button.
- Point out the **Live AI Prediction Card** showing classification, 98.5% confidence, and feature percentage reasoning ('+5,580% over baseline packet rate').

6. In the Security Events table, click **Investigate** on the DoS alert to display the 5-stage Alert Lifecycle, analyst guidance, and active iptables firewall blocking command.
7. Scroll to **Model Evaluation** section and present the measured 5x5 Flow Confusion Matrix and 2x2 ARP Confusion Matrix.

4. Examiner Q&A; Defense Script (Anticipated Questions & Answers)

Anticipated Examiner Question	Defensible Viva Response
Q1: What neural network model is actually implemented?	<i>'We implement a Deep Multi-Layer Perceptron (Deep MLP) with hidden layer structure (128, 64, 32), ReLU activation, and Adam optimizer using Scikit-Learn MLPClassifier. It processes 12 flow features extracted from network streams.'</i>
Q2: How does the system handle zero-day attacks?	<i>'We implement a Statistical Outlier Distance Engine. If traffic features deviate more than 3 standard deviations from all known baseline classes, the system flags the flow as Zero-Day / Novel Anomaly (MITRE T1204).'</i>
Q3: How does feature explainability work without black-box guessing?	<i>'The Ensemble Correlation Engine calculates exact feature percentage deltas relative to measured normal baselines (e.g. +5,100% over baseline packet rate of 50 pkts/s), grounding explanations strictly in empirical metrics.'</i>
Q4: Why use Random Forest for ARP Spoofing instead of Neural Net?	<i>'ARP protocol poisoning is characterized by discrete rule violations (unsolicited reply ratio, MAC binding conflicts). Random Forest decision trees handle tabular protocol threshold boundaries with 100% precision.'</i>